

Haoxiang You

 Haoxiang You |  [haoxiangyou.github.io](https://github.com/haoxiangyou) |  HaoxiangYou
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Education

Yale University PhD in Mechanical Engineering (Robotics Track), GPA: 4.0/4.0	Expected May 2027
University of Pennsylvania MS in Mechanical Engineering (Robotics Track), GPA: 4.0/4.0	May 2023
Ningbo University BS in Mechanical Engineering	Jun 2020
Purdue University Visiting Student	Dec 2018

Summary

Final-year Robotics PhD candidate at Yale with experience across the robotics stack — hardware design, control, data collection, simulation, model training, agentic workflows, and real-robot deployment. My research develops efficient robot learning algorithms. More recently, my interests lie in agentic robotics: evaluating coding agents on long-horizon physical tasks, integrating them as the robot brain, and using them as a scalable data engine for training low-level cerebellum policies such as VLAs. You can find some examples [here](#).

Publications

EmbodiedSWE: Coding Agents for Long-Horizon Dexterous Robotics <i>Haoxiang You, Z. Shen, Y. Liu, Z. Zheng, L. Zha, K. Yamazaki, M. Zhang, S. Huang, J. Sun, Q. Chen, L. He, H. Chang, D. Shah, M. Schwager, K. Fragkiadaki, P. Henderson, I. Abraham, C. Xu</i>	Under Review
Accelerating Visual-Policy Learning through Parallel Differentiable Simulation <i>Haoxiang You, Yilang Liu, Ian Abraham</i>	NeurIPS 2025 [Spotlight]
Efficient On-policy Visual-RL via Stochastic Decoupled Policy Gradient <i>Haoxiang You, Y. Liu, D. Zong, Qian Wang, T. Vitchutripop, Qi Wang, D. Rakita, I. Abraham</i>	CoRL 2026
Sample-Based Hybrid Mode Control: Asymptotically Optimal Switching of Algorithmic and Non-Differentiable Control Modes <i>Yilang Liu, Haoxiang You, Ian Abraham</i>	ICRA 2026
Goal-Driven Reward by Video Diffusion Models for Reinforcement Learning <i>Q. Wang, M. Wu, Y. Zhang, M. Yuan, W. Zhang, Haoxiang You, Y. Wang, X. Jin, X. Yang, W. Zeng</i>	CVPR 2026
Is Bellman Equation Enough for Learning Control? <i>Haoxiang You, Lekan Molu, Ian Abraham</i>	Preprint 2025

Work Experience

Genesis-AI , Research Scientist Intern – Built exoskeleton teleoperation hardware and a data-collection pipeline that gathers robot demonstrations for Vision-Language-Action (VLA) and imitation-learning training, and constructed digital twins of real scenes for simulation. – Developed a general-purpose Model Predictive Control (MPC) framework that generates manipulation data autonomously, without human teleoperation, and contributed training optimizations to the sim-to-real co-training pipeline.	Jul 2025 – Nov 2025
University of Pennsylvania , Robotics Research Assistant – Developed control strategies — trajectory optimization, feedback control, system identification, and state estimation — for the Cassie bipedal robot, and transferred locomotion controllers from simulation to hardware, bridging the sim-to-real gap under model mismatch, sensor noise, and actuator dynamics.	May 2022 – Sep 2022
YITU Tech , Self-Driving Control Intern – Trained neural vehicle-dynamics models to improve simulator fidelity for self-driving control, and tuned trajectory-tracking controllers on electric test vehicles.	Jan 2021 – Jul 2021

Selected Projects

Yale MCube Satellite

2024 – 2025

– Led the mechanical team — redesigning the satellite’s structure for manufacturing — and developed communication and control code as a core member of the embedded team.

Full-Stack LLM Implementation

2025

– Implemented an LLM training and deployment stack end-to-end — pretraining, supervised fine-tuning, reinforcement learning, and distributed training.

Invited Talks & Media

NVIDIA Robotics Seminar, Computational Methods for Curious Robots

2025

Robot Mind Podcast, Accelerating Reinforcement Learning with Visual Inputs

2025

Teaching & Academic Service

Reviewer, ICRA, CVPR, ECCV, NeurIPS, CoRL, RA-L

Organizer, Yale Robotics Reading Group

Teaching, Mechatronics Laboratory & Multivariable Calculus (Yale); Learning in Robotics (Penn)

Honors and Awards

Outstanding Academic Award	University of Pennsylvania	2023
Dean’s List	Purdue University	2018
Province Scholarship	Zhejiang	2017 & 2018
First-class Academic Scholarship	Ningbo University	2017

Skills

Programming	Python, C, C++, C#
Physical Simulation	MuJoCo, Brax, Isaac Sim, Newton, Genesis, ManiSkill, Drake
ML & Optimization	PyTorch, JAX, SNOPT, IPOPT
LLM Agents	Agent harnesses & evaluation, agentic coding workflows
Hardware & Tools	SolidWorks, mechatronics, 3D printing, CNC machining, Unity, Blender